Higher National Diploma in Quantity Surveying

2nd Year, First Semester Examination – 2023

QS 21054: Civil Engineering Construction II for QS

Instructions for Candidates:

Answer five questions only

All questions carry equal marks

No. of questions : 06
No. of pages : 06
Time : 03 hours

Q1:

i. What is the role of "Admixtures" in a concrete mixture. (02 Marks)

ii. Name two types Admixtures which are commonly used in the industry. (02 Marks)

iii. State three types of tests which determine the quality of aggregates. (03 Marks)

iv. Describe two methods of cement deterioration. (02 Marks)

v. Briefly explain the following Cement types and their applications in the industry.

a) Ordinary Portland cement

b) Portland Pozzolana cement

c) Rapid Hardening Cement

d) Low Heat Cement (02 x 4 Marks)

vi. Discuss the impacts of mixing several cement types together to prepare a concrete mixture for a slab construction. (03 Marks)

(Total 20 Marks)

Q2.

i. List down four structural systems which are used in a high-rise building. (04 marks)

ii. Name three challenges in constructing a high-rise building? (03 marks)

iii. Mention five components of HVAC systems (05 marks)

iv. Briefly discuss about two types of Air Conditioning system. (04 marks)

v. Explain two passive techniques used to control the thermal comfort within the building. (04 marks)

(Total 20 Marks)

Q3.

i. Classify different types of fire by providing relevant examples. (05 marks)

ii. Briefly explain the following in a fire protection system.

a) Fire Alarm

b) Foam based extinguisher (02x3 marks)

c) Fire sprinkler system (04 marks)

iii. Mention four advantages of Steel as a structural material. (02 marks)

iv. State four Load bearing components of a building. (03 marks)

v. Discuss about the Applicability of Precast concrete.

(Total 20 Marks)

Q4.

i. State five advantages of green buildings over the conventional building. (05 Marks)

ii. Mention four challenges in sustainable building construction? (04 Marks)

iii. Name three factors that a designer should consider in "Selecting Sanitary Appliances" for a multi-storey building. (03 Marks)

iv. Draw the "Solid waste management hierarchy" with relevant elements. (05 Marks)

v. Describe the importance of "Integration of building services" for a smooth functioning of a Hotel building. (03 Marks)

(Total 20 Marks)

Q5.

i. Mention 03 characteristics of a good formwork. (03 marks)

ii. Briefly explain 03 design consideration of the lighting system of a building. (06 marks)

iii. What is an "Earthing of the Electrical system". Why the building designers should consider about the Earthing system. (05 marks)

iv. State four basic components of a staircase. (02 marks)

v. List out four benefits of usage of Escalators in a high-rise building. (04 marks)

(Total 20 Marks)

Q6.

i. Define "Segregation and Bleeding" in fresh concrete. (02 mar

ii. Write the main steps of "Slump test" (03 mar

iii. Mention three qualities of a properly proportioned concrete mix. (03 mar

iv. Discuss the possible reasons for mis-matching the theoretical data and practical out used in the mix design calculation. (02 mar

v. Calculate the quantities for a trial mix which appropriates to a concrete mixture G 30, having workability of 30-60 mm in terms of slump test.

Used cement type is O.P.C. (strength class = 42.5). Maximum size of the crushed co aggregate is 20 mm. In addition, river sand was used as fine aggregate with estim percentage of passing 600μm sieve as 50 % from the total aggregate content. Rela density for combined aggregate density is 2.65.
Assume 2.5% defective and Standard Deviation as 4.0.
Maximum W/C ratio is 0.5 and minimum cement content is 350 kg/m^3 as per specification of BS 8110: Part I.
You may use following tables and graphs for calculations.

Clearly mention all the references and assumptions. (10 Mar

(Total 20 Mar

Table 01: probability factor

Defectives %	K value
10	1.28
5	1.64
2.5	1.96
1	2.33

Table 02 : Compressive strength of concrete mixes made with a free-water cement ratio of 0.5

Cement strength class	*Type of coarse aggregate*	*Compressive strength (N/mm²)*			
		Age (days)			
		3	*7*	*28*	*91*
42.5	Uncrushed	22	30	42	49
	Crushed	27	36	49	56
52.5	Uncrushed	29	37	48	54
	Crushed	34	43	55	61

Table 3: Approximate free water content required to give various levels of workability

Maximum size of aggregate (mm) (m)	Slump (mm)	0-10 > 12	10-30	30-60	60-180
	V-B (sec)		6-12	3-6	0-3
	Type of aggregates				
10(3/8)	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20(3/4)	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40(3/2)	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

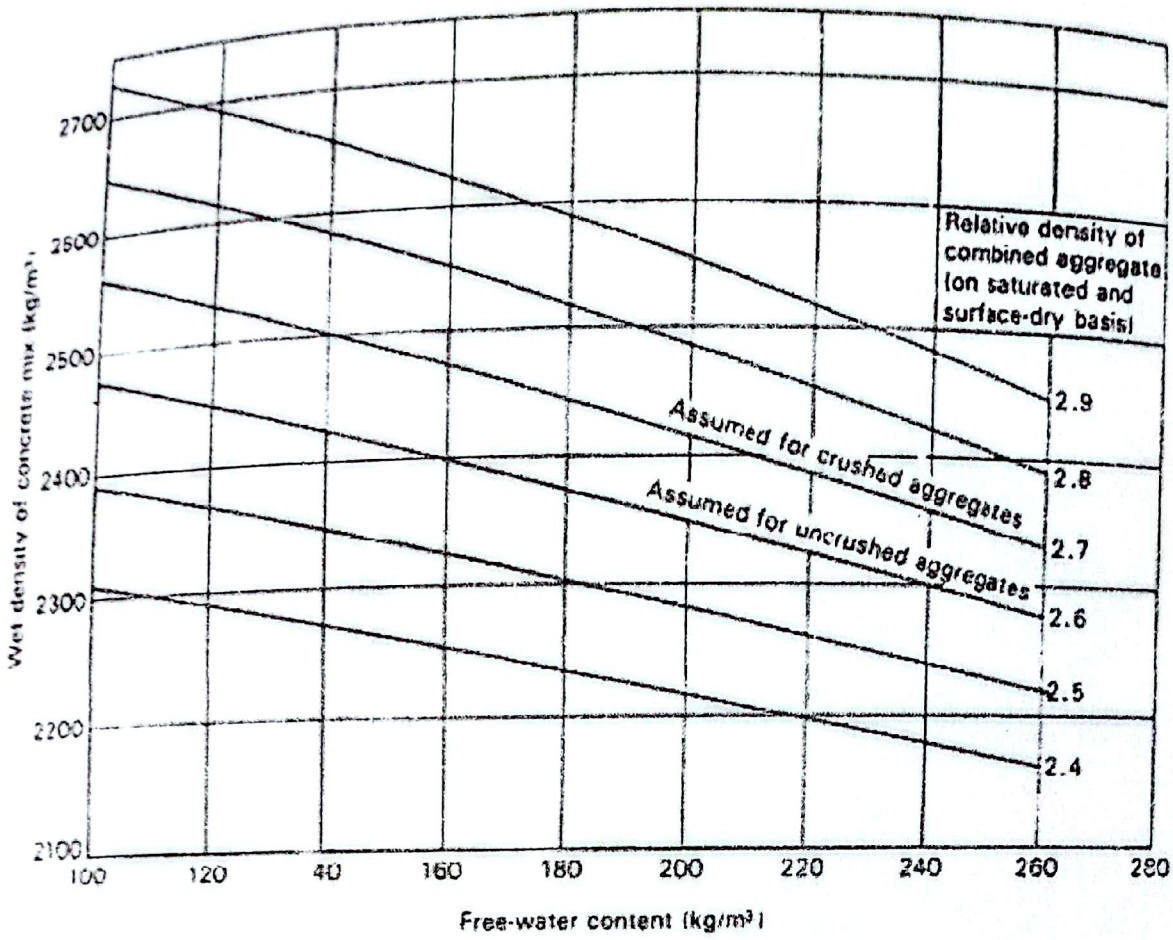

Figure 02: Estimated wet density for fully compacted concrete

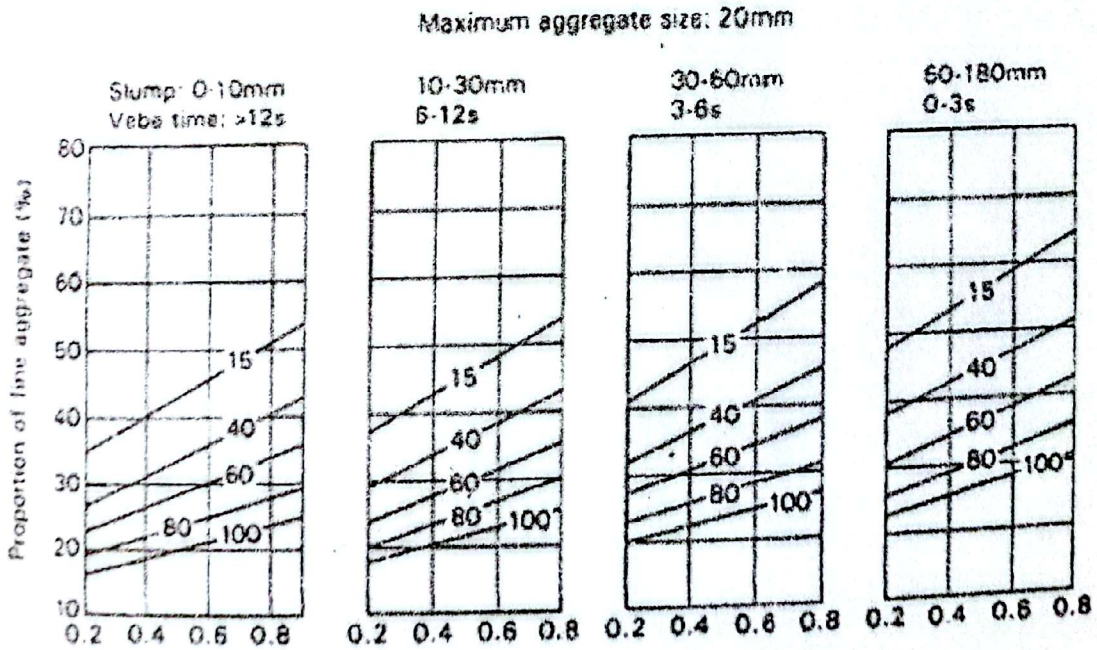

Figure 03: Recommended % of fine aggregate as a function of free w/c ratio for various values of workability and mix

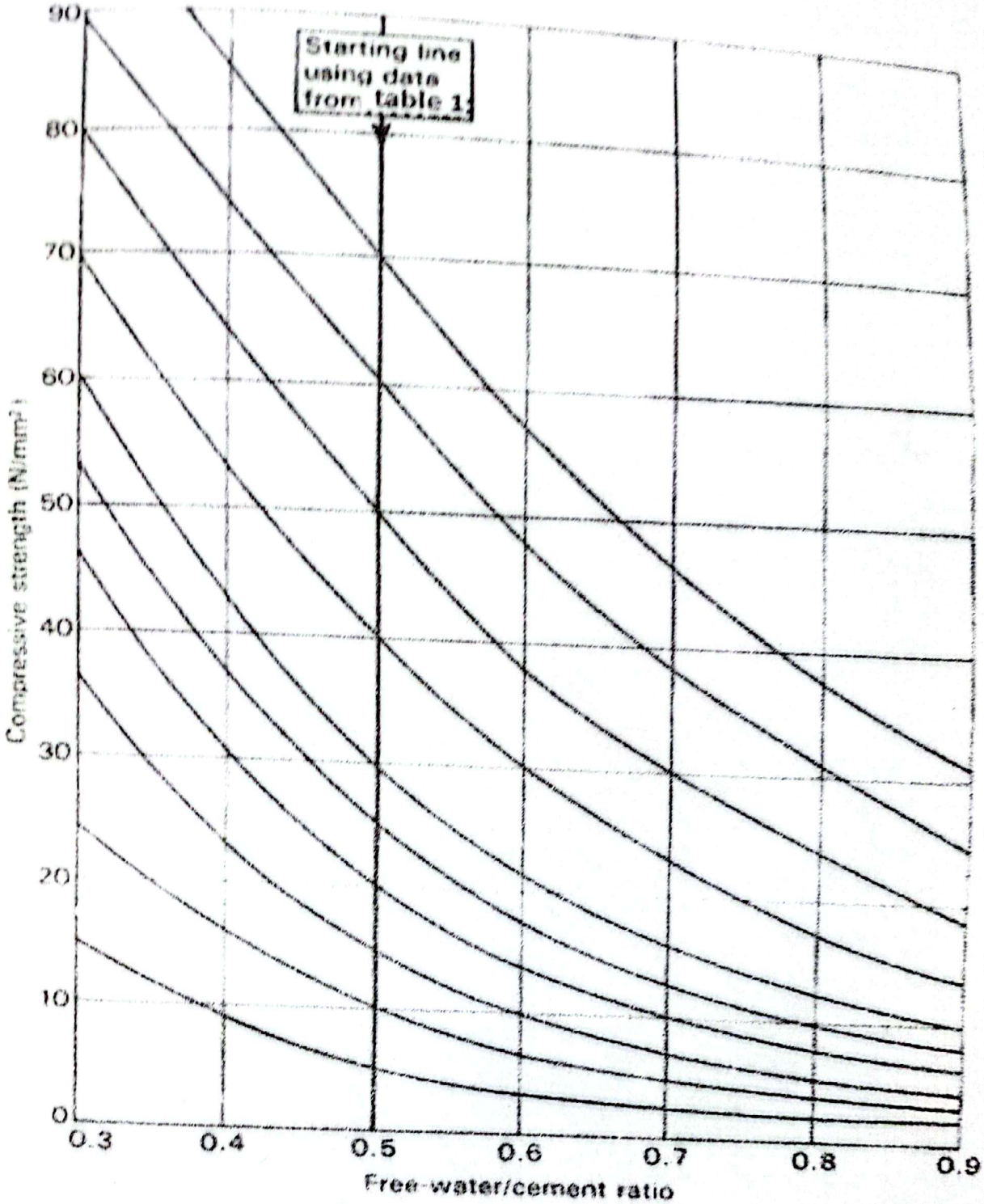

Figure 01: Relation between compressive strength and free w/c ratio